

STOR 155.010 Lecture 06

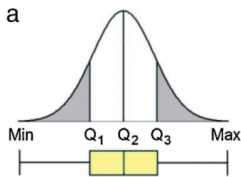
Correlation

Aniruddhan (Ani) Ganesaraman

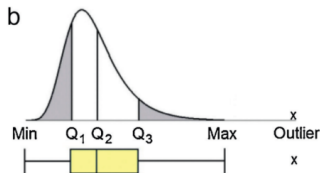
Fall 2026
UNC Chapel Hill

03 September 2026

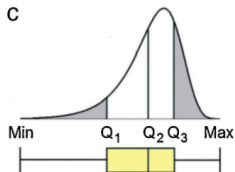
Boxplot and Skewness



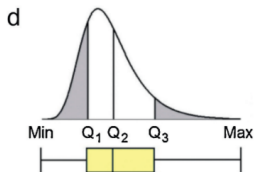
Symmetric



Asymmetric
(positive or right skewed)



Asymmetric
(negative or left skewed)



Asymmetric
(positive or right skewed)

Introduction

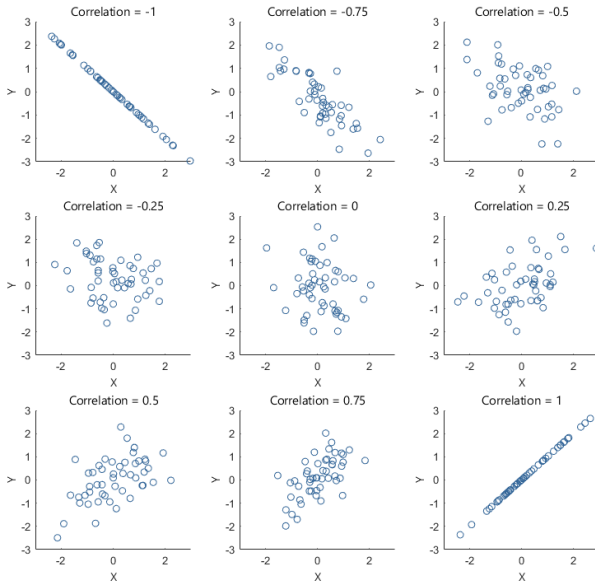
- Measures the strength and direction of a linear relationship between two numerical variables.
- Pearson's correlation coefficient:**

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right),$$

where $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ and $s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.

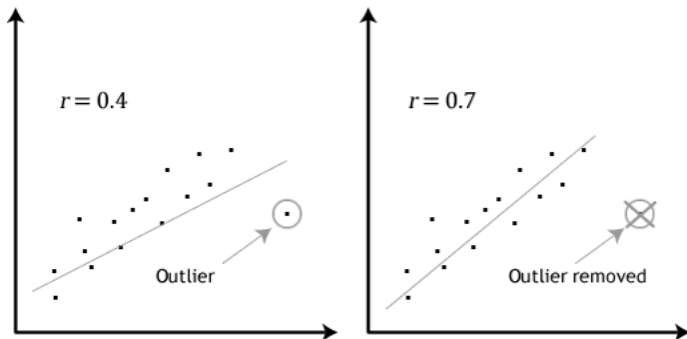
- Always between -1 (strong negative) and 1 (strong positive).
- A value close to 0 implies little to no linear correlation between the variables.

Scatterplots and Correlation



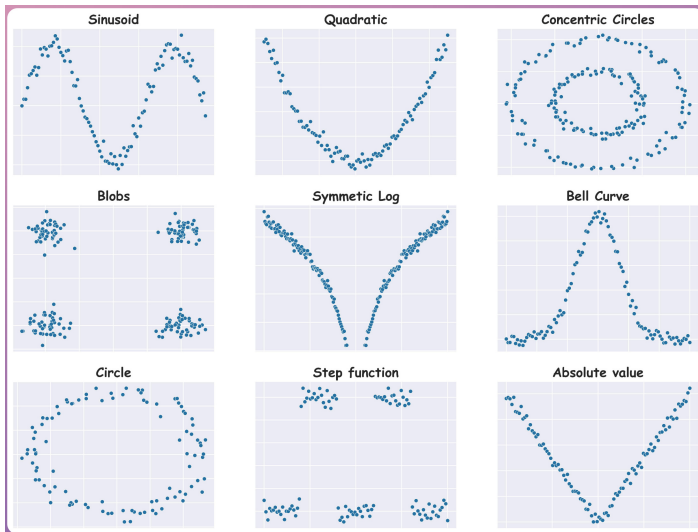
Correlation and Outliers

Correlation is NOT robust to outliers!



Datasets with **zero** correlation

Correlation only measures the linear relationship between two variables.



Limitations

- Correlation measures only linear relationships. Even strong non-linear relationships might have correlation close to zero.
- Correlation is not robust against outliers. A single outlier can significantly change r .
- Correlation does not imply causation. A third factor might influence both variables.
- **Tip.** Always visualize your data with scatterplots to complement the correlation analysis and spot outliers.

Example

- 5 data points: (2, 4), (4, 6), (6, 10), (8, 8), (10, 12).

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6, \quad \bar{y} = \frac{4 + 6 + 10 + 8 + 12}{5} = 8$$

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{40}{4} = 10, \quad s_x = \sqrt{10} \approx 3.16$$

$$s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 = \frac{40}{4} = 10, \quad s_y = \sqrt{10} \approx 3.16$$

Example (contd.)

x_i	y_i	$\bar{x}$	$\bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
2	4	6	8	16	16	16
4	6	6	8	4	4	4
6	10	6	8	0	4	0
8	8	6	8	4	0	0
10	12	6	8	16	16	16
Sum				40	40	36

$$r = \frac{1}{n-1} \cdot \frac{1}{s_x s_y} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{4} \cdot \frac{1}{\sqrt{10} \cdot \sqrt{10}} \cdot 36 = \frac{36}{40} = 0.9$$

References

- Section 8.1 of the textbook
- *Pearson's Correlation, Clearly Explained!!!* by StatQuest